MECHATRONICS SYSTEM INTEGRATION (MCTA 3203)

SEMESTER 1 2025/2026

PROGRESS REPORT #1

MICROCONTROLLER-BASED AUTOMATIC WASHING MACHINE WITH LOAD DETECTION

SECTION 2

GROUP 16

LECTURER: ZULKIFLI BIN ZAINAL ABIDIN & WAHJU SEDIONO

NO.	GROUP MEMBERS	MATRIC NO
1.	MUHAMMAD HAIKAL ZULHILMI BIN FAIROF	2313025
2.	MUHAMMAD FAIZ IZWAN BIN NOR EFFENDI	2313509
3.	AINUL JAARIAH BINTI ALIUDIN	2312664

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1. Introduction

The progress made during Week 10, which concentrated on creating the conceptual design for the small smart washing machine project, is shown in this weekly report. Before going into detailed design and implementation, this stage aims to precisely define the system concept, general working principle, and the interaction between hardware and software components.

2. Project Overview

The project involves designing and developing a **mini smart washing machine prototype** integrated with sensors and a microcontroller system. The system is intended to automate basic washing processes such as weight detection, water filling, washing, rinsing, and draining, while displaying real-time information on an LCD. Future enhancement includes the use of an **ESP32-CAM** for smart monitoring.

3. Conceptual Design Description

The conceptual design is based on a **microcontroller-controlled washing cycle**. The Arduino Uno acts as the main controller that receives input from sensors, processes data, and controls output devices such as motors, pumps, LEDs, and display modules. The washing process follows a predefined sequence to ensure proper operation.

3.1 System Concept

The conceptual design is based on a **microcontroller-controlled washing cycle**. The Arduino Uno acts as the main controller that receives input from sensors, processes data, and controls output devices such as motors, pumps, LEDs, and display modules. The washing process follows a predefined sequence to ensure proper operation.

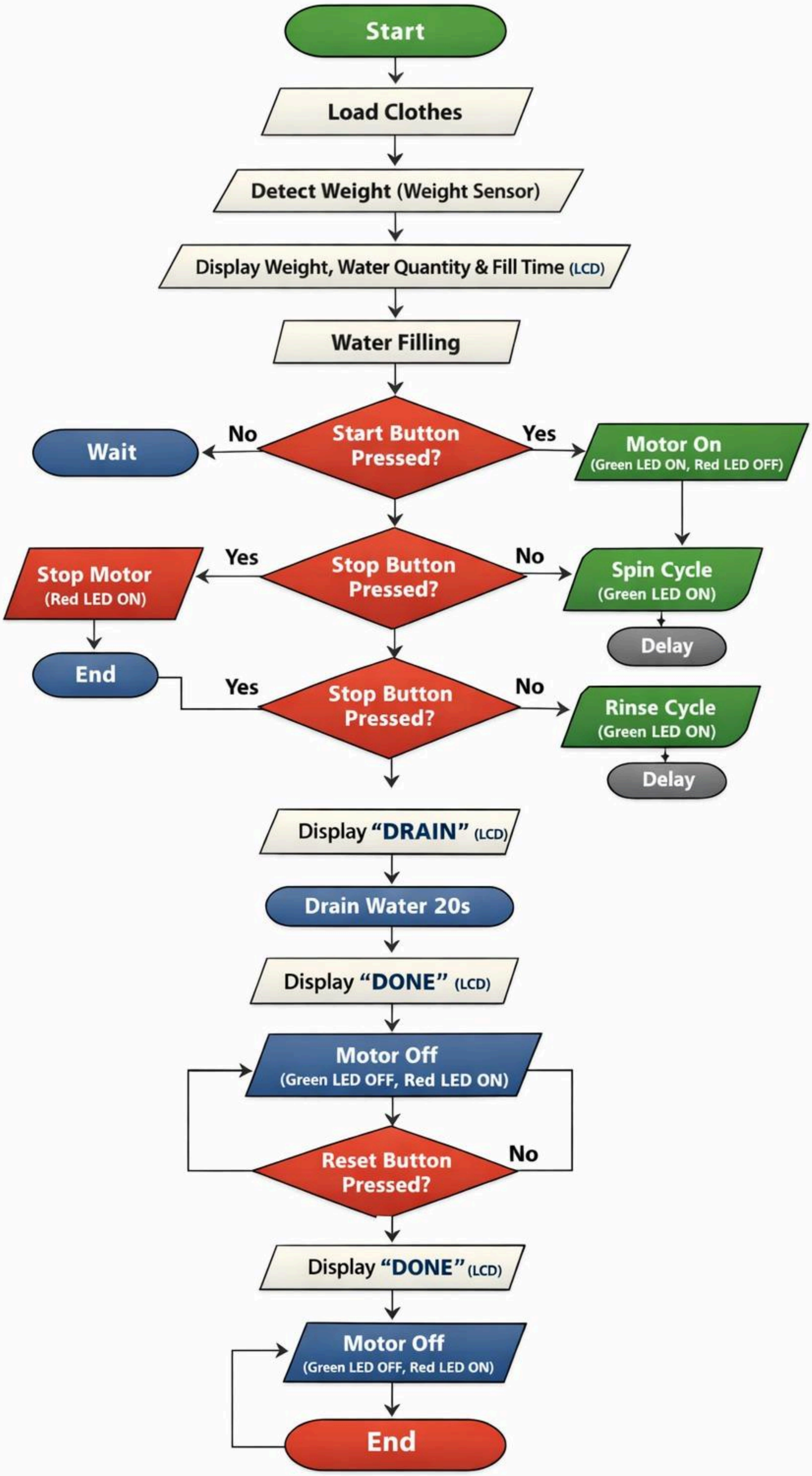
3.2 Main Components Identified

The following key components were selected and justified during this week:

1. **Arduino Uno** – Main controller for system logic and sequencing
2. **Weight Sensor with HX711 Module** – Measures the weight of clothes
3. **DC Motor** – Simulates drum rotation
4. **LCD (I2C)** – Displays process status and time
5. **LED Indicator** – Indicates motor operation status
6. **ESP32-CAM (Optional Enhancement)** – Enables visual monitoring and smart features

3.3 Process Flow Concept

1. User loads clothes into the drum
2. Weight sensor measures total load
3. Water is filled based on detected weight
4. Washing motor rotates for a fixed duration
5. Water is drained
6. Rinsing process begins
7. Final draining and process completion


Start
Load Clothes
Detect Weight (Weight Sensor)
Display Weight, Water Quantity & Fill Time (LCD)
Water Filling
Start Button Pressed?
No
Wait
Yes
Motor On
(Green LED ON, Red LED OFF)
Stop Button Pressed?
Yes
Stop Motor
(Red LED ON)
End
No
Spin Cycle
(Green LED ON)
Delay
Yes
Stop Button Pressed?
No
Rinse Cycle
(Green LED ON)
Delay
Display "DRAIN" (LCD)
Drain Water 20s
Display "DONE" (LCD)
Motor Off
(Green LED OFF, Red LED ON)
Reset Button Pressed?
No
Display "DONE" (LCD)
Motor Off
(Green LED OFF, Red LED ON)
End

4. Software Concept

At the conceptual level, the software design uses:

1. Sequential control logic
2. Fixed time-based operations (e.g., 20 seconds per process)
3. Real-time display updates on the LCD
4. Simple LED indicators for motor status

The ESP32-CAM is conceptually designed as a slave device that can later be integrated for image capture or IoT functionality without major changes to the main control logic.

5. Challenges Encountered

1. Ensuring the conceptual design remains simple while allowing future smart features
2. Deciding the level of ESP32-CAM integration without overcomplicating the system
3. Balancing hardware limitations with project objectives

<u>Certificate of Originality and Authenticity</u>

This is to certify that we are **responsible** for the work submitted in this report, that **the original work** is our own except as specified in the references and acknowledgment, and that the original work contained herein has not been untaken or done by unspecified sources or persons. We hereby certify that this report has **not been done by only one individual,** and **all of us have contributed to the report**. The length of contribution to the reports by each individual is noted within this certificate. We also hereby certify that we have **read** and **understand** the content of the total report, and no further improvement on the reports is needed from any of the individual contributors to the report. We therefore agreed unanimously that this report shall be submitted for **marking,** and this **final printed report** has been **verified by us.**

Signature: haikalzulhilmi

Name: Muhammad Haikal Zulhilmi Bin Fairof
Matric Number: 2313025

Read [/]
Understand [/]
Agree [/]

Signature: faizizwan

Name: Muhammd Faiz Izwan Bin Nor Effendi
Matric Number: 2313509

Read [/]
Understand [/]
Agree [/]

Signature:

Name: Ainul Jaariah Binti Aliudin
Matric Number: 2312664

Read [/]
Understand [/]
Agree [/]